**a Instrument detection**



Video frame $x_t$

640 px input setting

Backbone

Multiscale fusion

Detection heads

$P_3$

$P_4$

$P_5$

C

C

C

C

C = concatenation

box

class



Boxes, classes, scores

$s \geq 10^{-3}$ $\leq 100$ boxes

**b Question-driven template answering**

What type of forceps. . . ?

Question $q$

$r(q)$

Grammar router

polar

purpose

entity

Affirmative

Fixed purpose

Entity prior $e_f^*$

Training tool labels

Compose $T_{r(q)}$

The type of forceps. . .

Declarative response $\widehat{a}(q)$